

Treat the following exercises:

Exercise 1 (3 pts)

Given the following XML document **library.xml**:

```
<?xml version="1.0" encoding="iso-8859-1"?>
1<library>
2  <book>
3    <title>CSS 2</title>
4    <author>Daniel GLAZMAN</Author>
5    <editor>Eyrolles</editor>
6  </book>
7  <book>
8    Formation a... XML</title>
9    <author a=3>Michael J. YOUNG</author>
10   <editor>Microsoft Press</editor>
11 </book>
12 <book>
13   <title>Visual Basic .Net<author a="2" a="3">Richard CLARK</title></author>
14   <editor>Microsoft Press</editor>
15 </book>
16 <book>
17   <title>ASP.Net - To the .Point !</title>
18   <author a='4'>Grégory RENARD</author>
19   <editor>ASP-PHP.net</editor>
20 </book>
21</Library>
```

Is the document **library.xml** well formed? If not, for each error, indicate its line number and explain why and propose a correction.

Exercise 2 (2 pts)

Define an entity parameter *date* composed of day, month and year and use it in the following DTD:

```
<! ELEMENT fact (title, fact_date, info_date)>
<! ELEMENT title (#PCDATA)>
<! ELEMENT fact_date (#PCDATA)>
<! ELEMENT info_date (#PCDATA)>
```

Exercise 3 (3 pts)

Given the following XML document:

```
<AAA>
  <BBB/>
  <CCC/>
  <BBB/>
  <DDD>
    <BBB/>
    </DDD>
  <CCC>
    <DDD>
      <BBB/>
      <BBB/>
      </DDD>
    </CCC>
  </AAA>
```

Give the results of the following XPATH expressions:

- //DDD/BBB
- //CCC
- /AAA/BBB[LAST()]

Exercise 4 (4 pts)

Translate the following XML schema into a DTD:

```
<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
targetNamespace="http://www.journaux.com/">
<xsd:element name="journal" type="JournalType"/>
<xsd:complexType name="JournalType">
<xsd:sequence>
<xsd:element name="article" minOccurs="0" maxOccurs="unbounded">
<xsd:complexType>
<xsd:sequence>
<xsd:element name="titre" type="xsd:string"/>
<xsd:element name="pages">
<xsd:complexType>
<xsd:element name="page" minOccurs="0" maxOccurs="unbounded">
<xsd:simpleType>
<xsd:restriction base="xsd:int">
<xsd:maxExclusive value="100"/>
</xsd:restriction>
</xsd:simpleType>
</xsd:element>
</xsd:complexType>
</xsd:element>
<xsd:element name="auteur" type="xsd:string"/>
<xsd:element name="resume" type="xsd:string"/>
</xsd:sequence>
<xsd:attribute name="href" type="xsd:string"/>
</xsd:complexType>
</xsd:element>
</xsd:sequence>
<xsd:attribute name="title" type="xsd:string"/>
</xsd:complexType>
</xsd:schema>
```

Exercise 5 (4 pts)

Use the following DTD to build an XML document that will contain two elements CD. Use different cases in each of the CD elements and in function of a given data. Precise if you use a DTD interne or externe.

```
<!ELEMENT CDS(CD+)>
<!ELEMENT CD (Titre, Artiste+, Chanson*)>
<!ELEMENT Titre (#PCDATA)>
<!ELEMENT Artiste ((prenom*, nom) | groupe ?)>
<!ELEMENT prenom (#PCDATA)>
<!ELEMENT nom (#PCDATA)>
<!ELEMENT groupe (#PCDATA)>
<!ELEMENT Chanson (#PCDATA)>
<!ATTLIST Chanson genre CDATA # IMPLIED>
<!ATTLIST Chanson annee CDATA # IMPLIED>
<!ATTLIST Chanson duree CDATA # IMPLIED>
```

Exercise 6 (4 pts)

We wish to represent an address book in XML. For each entry of the book, we want to save the following information:

- name and first name of the person
- address (eventually outside the France)
- telephone number (eventually many)
- mail address (eventually many)
- birth date

a) Write a DTD for the address book.

b) Write a valid file for this DTD, containing at least two entries and makes appearing all the DTD possibilities (i.e. all the possible values for the attributes and all the optional elements and attributes).